CLASS 7 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A man goes from A to B at 4 km/h and returns at 6 km/h. The time taken in the round trip is 5 h. The distance AB is: 1

(a) 6 km **(b)** 5 km **(c)** 12 km **(d)** 24 km

2. In one hour this volume is: 1

(a) 8 litres **(b)** 480 litres **(c)** 4800 litres **(d)** 48 litres

3. Which of these statements about menarche is correct? 1

(a) it can occur from about 10 to 15 years **(b)** it occurs at 60 years
(c) it never occurs **(d)** it occurs at 5 years

4. Which of the following is a correct application of radiation? 1

(a) solar cooker **(b)** melting of ice **(c)** heating of milk **(d)** spoon in tea

5. A student measures one oscillation of a pendulum with a stopwatch of reading to 0.1 s and gets 2.0 s. The error is about: 1

(a) 50 per cent **(b)** up to 5 per cent
(c) up to 0.5 per cent **(d)** nil

6. A carnivore's digestive tract compared to a herbivore's is: 1

(a) same **(b)** absent **(c)** shorter **(d)** longer

7. A nurse says that the thyroid gland needs iodine. A diet lacking it may lead to: 1

(a) swelling of the neck **(b)** swelling of the foot
(c) swelling of the arm **(d)** swelling of the eye

8. A student places a thermometer in sunlight and another in shade. The one in sunlight reads higher due to: 1

(a) conduction **(b)** radiation **(c)** convection **(d)** evaporation

9. In 50 s he covers: 1

(a) 200 m **(b)** 100 m **(c)** 20 m **(d)** 400 m

10. The villi are absent in the intestine. The consequence is: 1

(a) no excretion **(b)** no breathing
(c) no digestion **(d)** poor absorption of digested food

11. If a person's pancreas stops making insulin, blood sugar will: 1

(a) rise **(b)** stay same
(c) become acidic always **(d)** fall to zero

12. In deserts, nights are cold and days are hot because sand: 1

(a) reflects all **(b)** heats slowly
(c) heats and cools quickly **(d)** stores heat

13. **Assertion (A):** The time period of a simple pendulum increases when its length is increased. 1
Reason (R): A longer pendulum takes more time to complete one oscillation.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** Bile, which is made by the liver, helps in the digestion of fats. 1
Reason (R): Bile contains enzymes that break fats into fatty acids.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** Testosterone is secreted by the testes. 1
Reason (R): Oestrogen is secreted by the ovaries.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why can we use speed = distance/time directly for uniform motion? 2

17. Why does oxygen move from the alveoli to blood? 2

18. Why do two persons of the same age show different features of puberty? 2

19. Why are the fins of a radiator painted black and made of metal? 2

20. A student recorded: 10 oscillations in 15.2 s, 15.0 s, 15.4 s. Find the mean time period. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A person goes from A to B at 40 km/h and returns at 60 km/h. Find the average speed. 3

22. A runner's pulse rate goes from 70 to 150 per minute. Explain what happens in the muscles and the blood. 3

23. Explain the importance of exercising in daylight for bones. 3

24. Explain how a refrigerator removes heat from inside to outside (in simple words). 3

OR

Explain the use of the 'seconds pendulum' in old clocks. 3

25. Why does a diabetic person need to be careful about sugar-rich foods? 3

OR

How would you respond if someone teases a classmate about body changes? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Kavya makes a simple pendulum using a stone tied to a 1 m thread. She counts 30 complete oscillations in 60 seconds. Later, she makes a longer pendulum and finds it takes 50 seconds for 20 oscillations. 5

(i) Find the time period of the first pendulum.

(ii) Does the time period depend on the mass of the bob?

(iii) Find the time period of the longer pendulum.

(iv) Which pendulum is longer, and how do you know?

27. Design an experiment to show how the time period of a pendulum varies with its length. Specify measurements, tables and expected relation. 5

OR

Describe the flow of blood from the heart through arteries, capillaries and veins and back, and say what is exchanged at the capillaries. 5

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND